

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

This month, in Earth we look at the problem of sinking coastal cities, in Health, we explore regenerative ageing methods, and in Space Age, we celebrate one year of the Ingenuity helicopter.

Source: University of York



Prehistoric fire art

In a study done on 15,000-year-old 50 engraved stones unearthed in France, it has been found that our ancestors were pretty skilled at creating art with firelight.

<https://digit.in/fireart>

WHAT'S NEW

Biggest of all

The Hubble Space Telescope has successfully determined the size of the largest comet nucleus ever seen by astronomers. Named C/2014 UN271 (Bernardinelli-Bernstein), this mammoth comet's nucleus is about 50 times larger than most known comets. The people need not worry about this comet, though. It has been travelling at about 22,000 miles per hour, moving from the solar system's edge towards our Sun for over a million years now. But, as per the calculations done by the researchers working on getting more information about it, the comet would not be able to get any closer than a billion miles from the Sun, which is about as far as Saturn is from the star around which our solar system revolves. The first records



Source: NASA

of this comet's presence date back to November 2010. Since then, the researchers kept a close eye on every detail of this comet, at least what was available to them. The main issue that came their way was differentiating between the cloud of dust and its solid nucleus. But the team of researchers was able to draw conclusions based on the pictures that the telescope took and concluded that the mass of this comet was about 500 trillion tons. <https://digit.in/cometnucleus>

Focus on the target

Researchers from the New York University have been able to determine the reason behind our ability to keep our eyes focused on a single object or point while we are moving towards or reaching for it. In their research titled, Modulation of inhibitory communication coordinates looking and reaching, published in the journal Nature, the team discussed their findings. While it may seem to be a relatively simple biological



process, it is a complex neurological process. After a series of experiments, the researchers, as written in the paper, found that "... neurons in the reach region of the posterior parietal cortex can inhibit neuronal firing

in the parietal saccade region to suppress eye movements and improve reach accuracy." This process of the eye focusing on the given object was termed gaze anchoring. It was also found that the brain waves that were involved in this process fell in the 15-25 Hz frequency range and are known as the beta waves. In our brain, it is these waves are responsible for ensuring organized neural firing. <https://digit.in/focus>



Do we come from space?

A team of researchers have successfully identified the presence of the five bases that store information in DNA and RNA in humans in space rocks that have fallen onto the surface of Earth. <https://digit.in/spacehumans>



Neptune's cooler than we expected

A study done by researchers from the University of Leicester has brought to light the unexpected changes in temperature that took place on the Neptune in the last twenty years, showing that Neptune is cooler than we had expected it to be. <https://digit.in/nepcool>



Reptiles in danger

Findings published in the research paper titled, A global reptile assessment highlights shared conservation needs of tetrapods, have shown that about one in five reptile species living on Earth face the danger of extinction. <https://digit.in/reptilesdanger>